Anisotropic expansion on $T^3 \times S^1$

Model specification and conditional predictions

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1. Spatial sections and metric

Let $\mathcal{M} = I \times T^3 \times S^1$, where I is an interval of comoving proper time. Each hypersurface Σ_t at fixed t is a compact flat four-manifold with topology $T^3 \times S^1$. Assume that the three torus directions have equal scale factor $a(t)$ and the circle has radius $b(t) > 0$. The unit-period external coordinates z^i and angular coordinate $y \sim y + 2\pi$ give

$$ds^2 = -dt^2 + a^2(t)\delta_{ij}dz^i dz^j + b^2(t)dy^2, \qquad V_4(t) = \int_{\Sigma_t} \sqrt{\det g_{\text{spatial}}}\, d^4z = 2\pi a^3 b.$$

Compactness here is a property of each spatial hypersurface. The model makes no compactness assumption on the time interval. The compact coordinate is part of the hypothesized geometry. Inferring it from a set of observations is a separate claim.

2. Stress and initial data

Use $c = \hbar = 1$, five-dimensional Einstein gravity with $\kappa > 0$, a vacuum density $\rho_\Lambda > 0$, and a source with $\rho_C = -C/b^5$, $p_{i,C} = -\rho_C$ in the torus directions and $p_{y,C} = 4\rho_C$, with $C > 0$. The vacuum pressure is $-\rho_\Lambda$ in all directions. The prescribed compact pressure follows from $p_{y,C} = -d(b\rho_C)/db$. This source is conserved on the homogeneous background and is phenomenological.

Write $H = \dot{a}/a$ and $K = \dot{b}/b$. The distinguished initial event has $H_* > 0$, $K_* = -3H_*$, $\kappa\rho_\Lambda = 18H_*^2/5$ and $\kappa C/b_*^5 = 48H_*^2/5$. The spatial four-volume is stationary through second order there. A finite interval of exactly constant volume is not a postulate of this candidate.

Research direction: Kirtus Justus. Model design and mathematical assistance: ChatGPT (OpenAI). Physical background: Chodos and Detweiler, dynamical compactification; Saharian and Mkhitaryan, compact vacuum stress.

3. Local volume evolution

The constraint is $3H^2 + 3HK = \kappa(\rho_\Lambda + \rho_C)$. With $\theta = 3H + K$, the spatial equations give

$$\dot{H} + \theta H = \kappa(2\rho_\Lambda/3 - \rho_C), \qquad \dot{K} + \theta K = \kappa(2\rho_\Lambda/3 + 4\rho_C).$$

At the designated event they imply $\dot{H}_* = 12H_*^2$, $\dot{K}_* = -36H_*^2$ and $\ddot{\theta}_* = -144H_*^3$. The constraint propagates under this smooth evolution. Let $x = \log(a/a_*)$ and $R = \log(V_4/V_{4,*})$. Expanding around the event gives

$$R(x) = -24x^3 + 270x^4 + O(x^5).$$

The solution decreases through a horizontal inflection. The two sides of the curve differ in magnitude at equal absolute x.

x	circle radius ratio b/b_*	$10^6 R(x)$
-0.010	1.03048244	27.08235
0	1	0
$+0.010$	0.97042458	-21.59197

The entries are calculated from the full homogeneous equations. The quartic approximation has an absolute error below 0.383 ppm in R across this interval.

A prespecified four-epoch contrast

Anchor the epoch independently or estimate it on a separate training set. Reserve the four external log-lengths $-r, -r/2, r/2, r$ with $r = 0.01$. The contrast

$$K_4 = \{R(-r) - 2R(-r/2) + 2R(r/2) - R(r)\}/6$$

lies between 6.10546 and 6.10547 ppm in the prescribed solution. The weights cancel every constant, linear or quadratic polynomial at these four coordinates. Inference from measured ratios requires simultaneous control of the epoch, calibration, propagation and source selection errors. The four values cannot determine an exact third derivative in a larger class of smooth histories.

4. Relative-volume observables

A periodic massless probe has compact momentum n/b. In the local adiabatic regime a neutral $+1, -1$ pair at rest has total energy $2/b$. A weak coupling to two massless zero-mode carriers can permit annihilation. For a matched two-body transition each outgoing carrier has energy $1/b$. A separate neutral rest pair of fixed mass m supplies the reference energy m per carrier.

The external torus quantizes outgoing momentum to $2\pi|\ell|/a$, $\ell \in \mathbb{Z}^3$. For each transition require an appropriate outgoing shell or a calibrated finite-bandwidth response. An interaction vertex alone fixes neither an event rate nor a resolved peak. All attempted preparations and failures enter a prospective comparison.

Two lines and a tensor response

Let $A = a_o/a_e$, $D = b_o/b_e$ at emission and reception. Assume identified co-emitted lines, matched propagation paths, comoving observers and independent local calibrations. The received energies are m/A and $1/(Ab_e)$. The ratio of the received lines divided by the locally calibrated $(1/b_o)/m$ gives D; the reference gives A. Thus $A^3D = V_{4,o}/V_{4,e}$ under the stated response model.

For a planar short-wave transverse tensor zero mode with negligible tensor anisotropic stress, the kinetic coefficient is proportional to a^3b and the instantaneous frequency is k/a. Wave action gives the leading strain ratio $T = A^{-1}D^{-1/2}$. When source strain and the line standards are independently calibrated, this tests $A^2DT^2 = 1$. A localized astronomical source also requires generation, distance and detector corrections.

If the reference mass changes by $g = m_o/m_e$, the inferred ratios are $A_m = gA$ and $D_m = gD$. With separately justified tensor transfer one can estimate $g = (A_m^2 D_m T^2)^{1/3}$. Using the tensor measurement for this calibration consumes it as an independent test of the transfer relation.

Tensor propagation context: Belgacem et al., Phys. Rev. D 97, 104066.

5. Resolved spectral levels

A fixed external Fourier index **k** gives per-particle energy

$$E^2_{n,\mathbf{k}} = (2\pi|\mathbf{k}|/a)^2 + n^2/b^2.$$

Normalize the energy and the individual external momentum magnitude by an independently identified first rest line $E_1 = 1/b$, giving $Q = E/E_1$ and $P = p/E_1$. The combination $Z = Q^2 - P^2$ is 1 for the first level and 4 for the second. A neutral pair has zero total momentum in either assignment, so resolution of its individual momentum is essential.

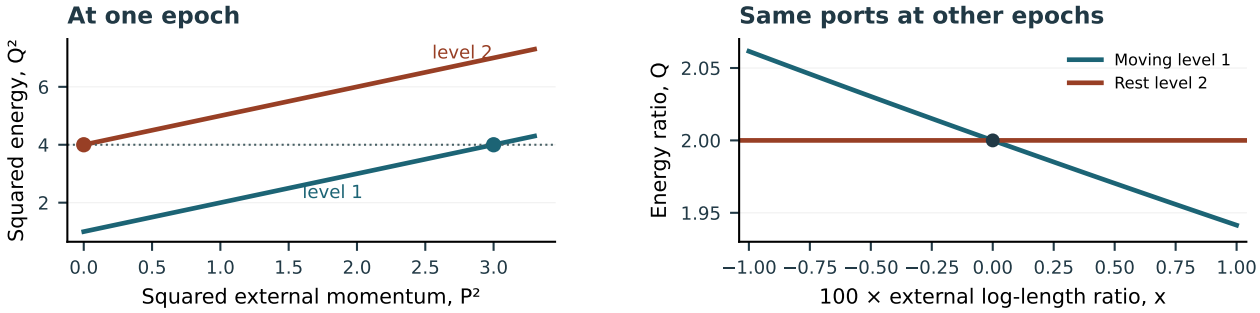


Figure 1. At one epoch a moving first-level pair and a second-level rest pair can have the same energy. At fixed external index their relative frequencies subsequently separate. Lines are leading adiabatic predictions for specified modes.

For a nonzero first-level external index define $\lambda_k = (2\pi|\mathbf{k}|b_*/a_*)^2$. The two assignments give $Q_1^2 - 1 = \lambda_k \exp(2R - 8x)$ and $Q_2^2 - 1 = 3$. At $x = \pm 0.01$ the predicted change from the later to the earlier value of $\log(Q^2 - 1)$ is in $[0.16009734862, 0.16009734864]$ for the moving first level and is zero for the second rest level. The comparison requires tracking the same nonzero external index, complete preparation records and a resolved line at both epochs.

These predictions distinguish two assignments within a specified spectrum. A determination of spatial topology would additionally require a measurement that separates the compact model from four-dimensional theories admitting the same masses, couplings and homogeneous evolution.

Compact Fourier spectrum: TASI lectures on extra dimensions.